FACULTY OF COMPUTING
UTM Johor Bahru

SCST1123: MATHEMATICS FOR SOFTWARE ENGINEERING
20242025/SEMESTER 1

ASSIGNMENT #4a (Chapter 8 & 9)

INSTRUCTION:

1. This is an assignment for a **GROUP of 3 students**. Please clearly write the course details including the group members name & matric number in the front page of the submission.
2. This assignment will give **5%** of overall course marks.
3. The submission must be neatly written (using MS Word or handwriting on a tablet). Save your work in PDF format and upload the file to e-learning before the deadline (**Monday - 20/01/2025, 5:00 PM)**
4. Any submission that is unreadable, will be **REJECTED**.

Question 1 **[10 Marks]**

Let the alphabet $\sum = \{x, y\}$

a) Define a language L_1 over Σ to be the set of all strings that begin with the character x and have length at most three characters. Find L_1. (2 marks)

b) Define a language L_2 over Σ to be the set of all palindromes obtained using the characters of Σ. Write ten elements of L_2. (4 marks)

c) Let $A = \Sigma^0 \cup \Sigma^1$ and $B = \Sigma^2 \cup \Sigma^3$. Use words to describe A, B and $A \cup B$. (3 marks)

d) Find Σ^3. (1 mark)

Question 2 **[10 Marks]**

a) Let $\sum = \{a, b\}$

i) Use words to describe the languages defined by the following regular expression:

$$a^*b^*|a^*b^*$$ (3 marks)

ii) Write five strings that belong to the language defined by the given regular expression.

$$0^*1(0^*1^*)^*$$ (2 mark)

b) Write the regular expression to define the given set of strings.

i) All words that are written in lower-case letters and start with the letters *pre* but do not consist of *pre* all by itself. (2 marks)

ii) All words that are written in lower-case letters and contain at least one of the vowels a, e, i, o, u. (3 marks)

Question 3 **[15 Marks]**

a) Given a deterministic finite automaton (DFA) as in Figure 1.

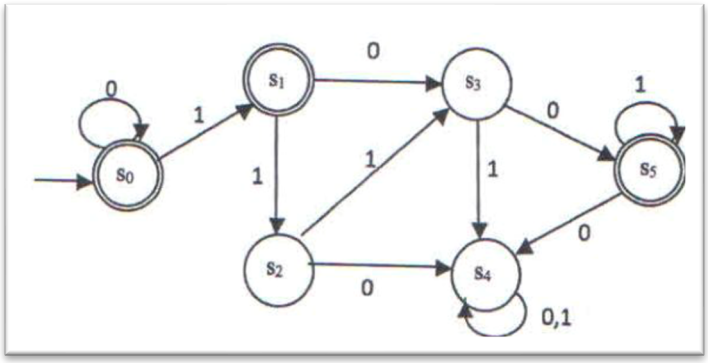


Figure 1

i) Write the set of final states for the DFA. (1.5 marks)

ii) Construct a transition table for the DFA. (3 marks)

iii) Given a string: **0011101100.** Find the sequence of configurations and state if the string is accepted or not by the DFA. (2.5 marks)

b) Construct a deterministic finite automaton (DFA) that accepts the set of all bit strings that contain three consecutive 1s. (3 marks)

c) Design a transition diagram of DFA that will accept the word '**banana**' whilst using only 3 states (e.g., S1, S2 and S3). (5 marks)

Question 4 **[15 Marks]**

a) Construct the state transition table for the finite state machine (FSM) defined by the transition diagram shown in Figure 2. Then find the output string for the input string **101011**.

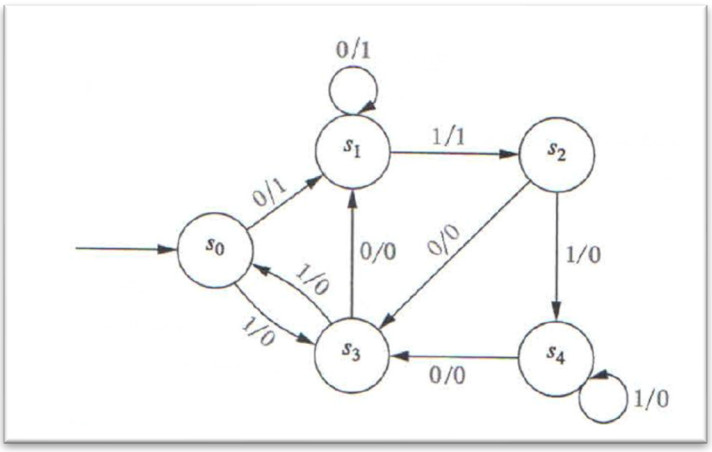


Figure 2

(7 marks)

b) A two-floor supermarket has provides an elevator for the convenient of their customers while shopping. The elevator can be at one of these two floors: ground floor or first floor. There is one button that controls the elevator, and it has two values: Up or Down. Also, there are two lights in the elevator that indicate the current floor: Red for Ground floor and Green for First floor. At each time step, the controller checks the current floor and current input, changes floors and light in the obvious way.

i) Construct the state transition table based on the following information.

States: Ground and First

Inputs: Up and Down

Outputs: Red or Green (4 marks)

ii) Draw the state transition diagram for this elevator mechanism. (4 marks)